

Program 3: Maven Project To Connect To Web Servers Using Selenium Dependency (Sauce.demo)

Create a Maven Project:

`mvn archetype: generate -DgroupId=com.example -DartifactId=MyMavenGuavaApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false`

```
rakshith@rakshith-VirtualBox:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/rakshith/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp01
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/rakshith/devops/MyMavenSeleniumApp01
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 2.217 s
[INFO] Finished at: 2026-03-19T14:17:33+05:30
[INFO]
rakshith@rakshith-VirtualBox:~/devops$
```

Before connecting to web we have to make sure that the browser which is being used has its driver installed

Downloading Chrome Driver:

To check the chrome version type the command `google-chrome --version` in the terminal

extract the version and store it in a variable `CHROME_VERSION`

extracts and transforms JSON data

```
LATEST_DRIVER=$(curl -s "https://googlechromelabs.github.io/chrome-for-testing/latest-patch-versions-per-build.json" | jq -r ".builds[\"$CHROME_VERSION\"].version")
```

do install curl using the command

```
sudo apt install curl
```

```
wget https://storage.googleapis.com/chrome-for-testing-public/$LATEST_DRIVER/linux64/chromedriver-linux64.zip
```

Extract and move it to `/usr/local/bin/`:

```
unzip chromedriver-linux64.zip
```

```
sudo mv chromedriver-linux64/chromedriver /usr/local/bin/chromedriver
```

```
sudo chmod +x /usr/local/bin/chromedriver
```

Check if ChromeDriver is installed correctly:

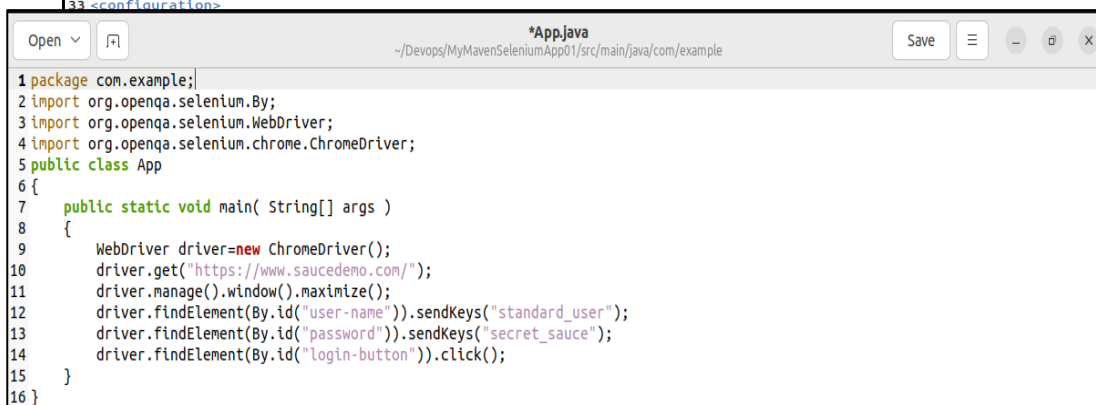
```
rakshith@rakshith-VirtualBox:~/devops$ chromedriver --version
ChromeDriver 145.0.7632.117 (de4e5d6e13b995f5e04139708ec456017467-refs/branch-heads/7632@{#3101})
rakshith@rakshith-VirtualBox:~/devops$ █
```

Open pom.xml and add all the required dependencies:



```
1 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
2   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
3   <modelVersion>4.0.0</modelVersion>
4   <groupId>com.example</groupId>
5   <artifactId>MyMavenSeleniumApp01</artifactId>
6   <packaging>jar</packaging>
7   <version>1.0-SNAPSHOT</version>
8   <name>MyMavenSeleniumApp01</name>
9   <url>http://maven.apache.org</url>
10  <dependencies>
11    <dependency>
12      <groupId>junit</groupId>
13      <artifactId>junit</artifactId>
14      <version>3.8.1</version>
15      <scope>test</scope>
16    </dependency>
17    <!-- Source: https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
18  </dependencies>
19  <dependency>
20    <groupId>org.seleniumhq.selenium</groupId>
21    <artifactId>selenium-java</artifactId>
22    <version>4.41.0</version>
23    <scope>compile</scope>
24  </dependency>
25  <!-- Build: Configuring plugins and build settings -->
26  <build>
27    <!-- Example: Maven Compiler Plugin to compile Java code -->
28    <!--
29    <plugin>
30      <groupId>org.apache.maven.plugins</groupId>
31      <artifactId>maven-compiler-plugin</artifactId>
32      <version>3.8.1</version>
33    </plugin>
34  </build>
35  </project>
```

Edit



```
1 package com.example;
2 import org.openqa.selenium.By;
3 import org.openqa.selenium.WebDriver;
4 import org.openqa.selenium.chrome.ChromeDriver;
5 public class App
6 {
7     public static void main( String[] args )
8     {
9         WebDriver driver=new ChromeDriver();
10        driver.get("https://www.saucedemo.com/");
11        driver.manage().window().maximize();
12        driver.findElement(By.id("user-name")).sendKeys("standard_user");
13        driver.findElement(By.id("password")).sendKeys("secret_sauce");
14        driver.findElement(By.id("login-button")).click();
15    }
16 }
```

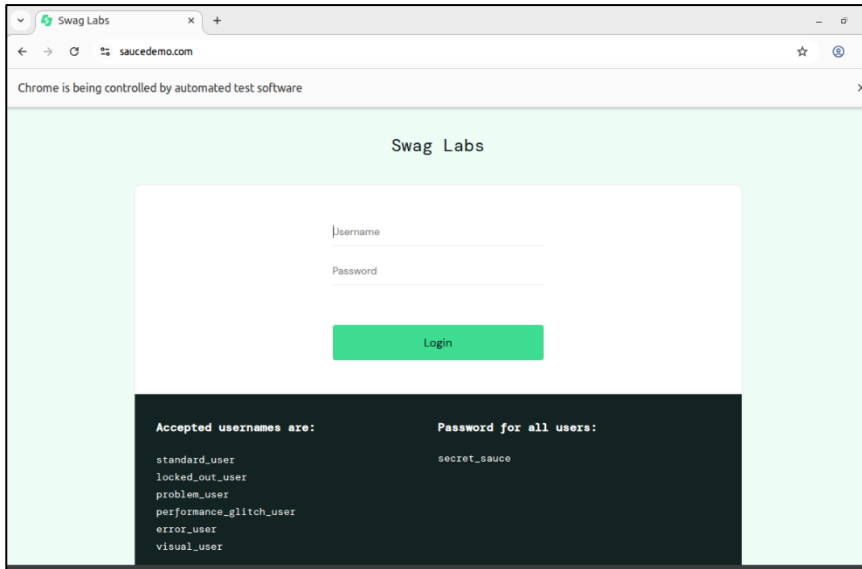
App.java:

Build the project using the command: mvn clean install

```
rakshith@rakshith-VirtualBox: ~/devops/MyMavenSeleniumApp01
[INFO] Scanning for projects...
[INFO] -----< con.example:MyMavenSeleniumApp01 >-----
[INFO] Building MyMavenSeleniumApp01 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenSeleniumApp01 ---
[INFO] Deleting /home/rakshith/devops/MyMavenSeleniumApp01/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenSeleniumApp01 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/rakshith/devops/MyMavenSeleniumApp01/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenSeleniumApp01 ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/rakshith/devops/MyMavenSeleniumApp01/target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenSeleniumApp01 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/rakshith/devops/MyMavenSeleniumApp01/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenSeleniumApp01 ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/rakshith/devops/MyMavenSeleniumApp01/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenSeleniumApp01 ---
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running con.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.01 s - in con.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- maven-jar-plugin:3.0.1:jar (default-jar) @ MyMavenSeleniumApp01 ---
[INFO] Building jar: /home/rakshith/devops/MyMavenSeleniumApp01/target/MyMavenSeleniumApp01-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenSeleniumApp01 ---
[INFO] Installing /home/rakshith/devops/MyMavenSeleniumApp01/target/MyMavenSeleniumApp01-1.0-SNAPSHOT.jar to /home/rakshith/.m2/repository/con/example/MyMavenSeleniumApp01/1.0-SNAPSHOT/MyMavenSeleniumApp01-1.0-SNAPSHOT.jar
[INFO] Installing /home/rakshith/devops/MyMavenSeleniumApp01/pom.xml to /home/rakshith/.m2/repository/con/example/MyMavenSeleniumApp01/1.0-SNAPSHOT/MyMavenSeleniumApp01-1.0-SNAPSHOT.pom
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 3.900 s
[INFO] Finished at: 2026-03-19T14:22:18+05:30
[INFO]
rakshith@rakshith-VirtualBox:~/devops/MyMavenSeleniumApp01$
```

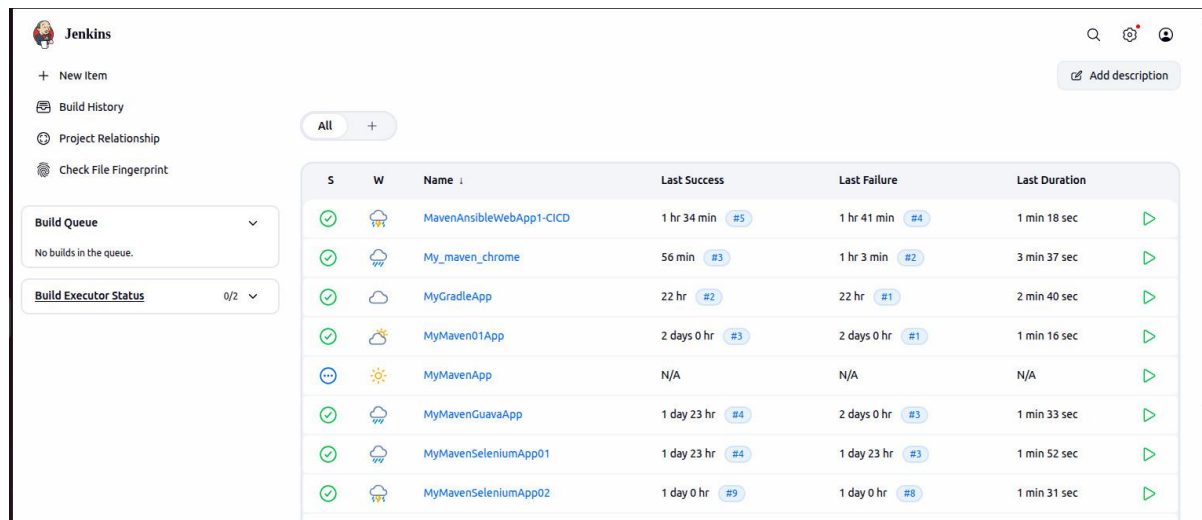
Now Run the command using

`mvn exec:java -Dexec.mainClass="com.example.App"`



Jenkins Pipeline for MyMavenSeleniumApp01:

Step 1:Open Jenkins Dashboard to create a new item



The screenshot shows the Jenkins Dashboard. On the left, there's a sidebar with navigation links: '+ New Item', 'Build History', 'Project Relationship', and 'Check File Fingerprint'. Below these are two panels: 'Build Queue' (showing 'No builds in the queue.') and 'Build Executor Status' (showing '0/2'). The main area displays a table of builds with columns: 'S' (Status), 'W' (Icon), 'Name', 'Last Success', 'Last Failure', and 'Last Duration'. The table lists several builds, including 'MavenAnsibleWebApp1-CICD', 'My_maven_chrome', 'MyGradleApp', 'MyMaven01App', 'MyMavenApp', 'MyMavenGuavaApp', 'MyMavenSeleniumApp01', and 'MyMavenSeleniumApp02'. The 'MyMavenApp' build is highlighted with a blue background.

S	W	Name	Last Success	Last Failure	Last Duration
✓	☁	MavenAnsibleWebApp1-CICD	1 hr 34 min #5	1 hr 41 min #4	1 min 18 sec
✓	☁	My_maven_chrome	56 min #3	1 hr 3 min #2	3 min 37 sec
✓	☁	MyGradleApp	22 hr #2	22 hr #1	2 min 40 sec
✓	☁	MyMaven01App	2 days 0 hr #3	2 days 0 hr #1	1 min 16 sec
...	☀	MyMavenApp	N/A	N/A	N/A
✓	☁	MyMavenGuavaApp	1 day 23 hr #4	2 days 0 hr #3	1 min 33 sec
✓	☁	MyMavenSeleniumApp01	1 day 23 hr #4	1 day 23 hr #3	1 min 52 sec
✓	☁	MyMavenSeleniumApp02	1 day 0 hr #3	1 day 0 hr #8	1 min 31 sec

Step 2: Click on New Item in order for creating an New Item and thus give the name as “MyMavenSeleniumApp01” and Item type as Pipeline and Create it

New Item

Enter an item name

MyMavenSeleniumApp01

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Maven project

Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

OK

Step 3: Configure the settings by Defining the script as “Pipeline Script from SCM” and SCM as “Git”. Thus, now as we have selected Git, we need to paste the Repository URL of the project given in GitHub and choose Credentials as “GitHub Credentials” which was created earlier.

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM

Git

Repositories

Repository URL

https://github.com/rakshith31-hub/MyMavenSeleniumApp01.git

Credentials

GitHub Credential

+ Add

Advanced

Save Apply

Step 4: Now in Terminal, go to directory created for MyMavenSeleniumApp01 and create a new file called “Jenkinsfile”

Paste the code inside Jenkinsfile

Enter following code into jenkins file

```
pipeline
{
    agent any // Use any available agent

    tools {
        maven 'Maven' // Ensure this matches the name configured in Jenkins
    }

    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url:
'https://github.com/rakshith31-hub/Mymavenapp.git'
            }
        }
    }
}
```

}

```
stage('Build')
{
    steps {
        sh 'mvn clean package' // Run Maven build
    }
}

stage('Test')
{
    steps {
        sh 'mvn test' // Run unit tests
    }
}

stage('Run Application') {
    steps {
        // Start the JAR application
        sh 'java -jar target/MyMavenApp-1.0-SNAPSHOT.jar'
    }
}

}

post {
    success {
        echo 'Build and deployment successful!'
    }

    failure {
        echo 'Build failed!'
    }
}
}
```

Step 5: We need to update pom.xml also in order to change the versions of source and target from 1.7 to 1.8

```
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>
```

Step 6: Now we need to Push changes and commit by pushing the necessary “Jenkinsfile” to the respective Repo

The screenshot shows the GitHub interface for a repository named 'MyMavenSeleniumApp01' by user 'rakshith31-hub'. The repository is public and has 0 stars, 0 forks, and 0 watchers. The main branch is 'main'. The file list shows a commit titled 'Update pom.xml' from 2 days ago, which includes files 'src', 'target', 'JenkinFile', 'Jenkinsfile', and 'pom.xml'. The 'About' section on the right indicates no description, website, or topics are provided. The 'Releases' and 'Packages' sections also show no published items.

Step 7: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.

Jenkins

All

MyMavenSeleniumApp01

Search

Settings

Help

Status

Changes

Build Now

Configure

Delete Pipeline

Full Stage View

Stages

Rename

Pipeline Syntax

Builds >

Filter

13 April 2026

#4 15:06

#3 14:50

#2 14:49

#1 14:47

MyMavenSeleniumApp01

Add description

Stage View

Average stage times:
(full run time: ~1min 52s)

	Declarative: Checkout SCM	Declarative: Tool Install	Checkout	Build	Test	Declarative: Post Actions
#5 Apr 13 15:06 1 commit	2s	635ms	2s	1min 1s	27s	287ms
#3 Apr 13 14:50 No Changes	2s	528ms	2s	15s failed	222ms failed	251ms
#2 Apr 13 14:49 No Changes						
#1 Apr 13 14:47 No Changes						

Permalinks

Last build (#4). 2 days 2 hr ago